
MACHINE LEARNING COURSEWORK: MIND READING

January 28, 2025

1 Comprehensive Dataset Exploration

1.1 KNOW

Type of Data	Number of Samples	Number of Features
Seen Brain	16540	561
Seen Image	16540	100
Seen Text	16540	512
Seen Label	16540	1
Unseen Brain	16000	561
Unseen Image	16000	100
Unseen Text	16000	512
Unseen Label	16000	1

Figure 1: Table of dataset shape.

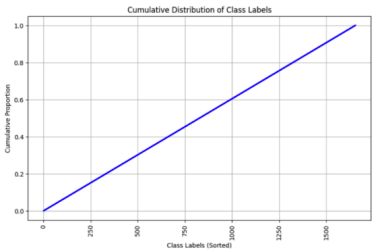


Figure 2: Cumulative class distribution graph.

1.2 SEE

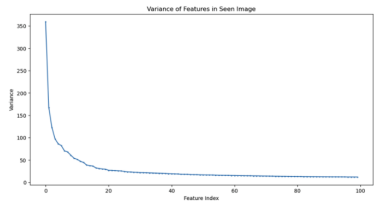


Figure 3: Image feature-wise variance graph.

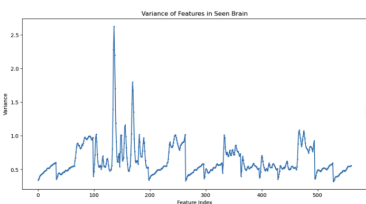


Figure 4: Brain feature-wise variance graph.

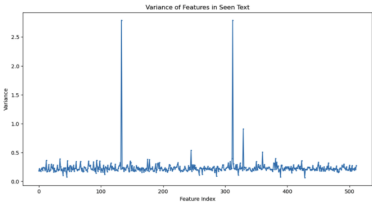


Figure 5: Text feature-wise variance graph.

Paragraph

1.3 FIND

The BraVL dataset displays a balanced class distribution, with each label associated with 10 samples across the datasets, ensuring consistency and linearity (Figure 1). Feature-wise variance analysis shows that some features exhibit high variance, signalling significant separation between values for certain features (Figure 1). This pattern is consistent across brain, image and text data (Figures 4 and 5), suggesting distinct clusters that can be effectively captured by distance-based algorithms like KNN. However, potential issues such as high-dimensional data, and noisy features could impact model performance. The text has shown that there are a large number of features with similar variances,

suggesting less informative features. Whereas image has shown an exponential decay pattern, suggesting that the number of informative features decreases exponentially. Visual evidence, including the cumulative distribution of classes (Figure 1) and feature-wise variance plots (Figures 4 and 5), supports these findings and underscores the importance of maintaining balanced class distributions and selecting relevant features for optimal model performance.

2 Custom Model Implementation

2.1 IMPLEMENT

For the classification task of labelling the data, we can use K-Nearest Neighbours (KNN) due to the multidimensional nature of the dataset (Figure 1). The feature-wise variance in the seen brain data indicates a variety of feature importance, with some features peaking showing high variance compared to the majority of features. This suggests that there is significant separation between values for certain features, making distance-based algorithms like KNN appropriate. The presence of high variance in features implies that the data points are spread out in the feature space, which is conducive to the effective use of distance metrics. This observation is also supported by the feature-wise variance in the image (Figure 4) and text (Figure 5) data, where similar patterns of variance are observed. The variety in feature values across different categories indicates that the data has distinct clusters, which can be effectively captured by KNN's reliance on proximity. Therefore, the use of a distance-based algorithm like KNN is justified for this classification task. The dataset exhibits a balanced distribution (figure 2), which supports the use of KNN. Balanced classes ensure that the algorithm's decision boundaries are not biased towards any particular class, allowing KNN to effectively capture the local structure of the data.

2.2 COMPARE

Sklearn KNN Model	
Performance Measure	Result
Accuracy	87%
F1-Score	83%
Recall	87%
Precision	84%

Figure 6: SKlearn KNN model results.

Implemented KNN Model	
Performance Measure	Result
Accuracy	85%
F1-Score	83%
Recall	85%
Precision	87%

Figure 7: Implemented KNN model results.

Both KNN models achieve very comparable performance overall, with each having its own strengths. The SKLearn-based KNN model shows a slightly higher accuracy (87% vs. 85%) and recall (87% vs. 85%), indicating that it correctly labels a marginally higher proportion of positive instances. On the other hand, the custom-implemented KNN model predicts a positive, it is correct more often. In practice, these differences are relatively small and could be influenced by factors such as parameter tuning or subtle differences in implementation details.

2.3 IMPROVE

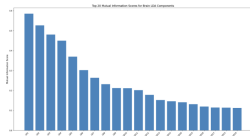


Figure 8: Top 20 Brain LDA components.

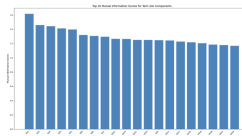


Figure 9: Top 20 Text LDA components.

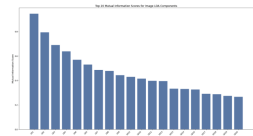


Figure 10: Top 20 Image LDA components.

To improve the model, refining the preprocessing of the data can improve the model. This is done via applying Linear discriminant analysis (LDA) to the brain, image and text data separately. Doing this before the merging of data is a feature reduction and transformation strategy. LDA will maximise the separation between the classes making features more informative for classification. As KNN is a distance based method, reducing features and therefore dimensions

will lessen the power of the curse of dimensionality. In high dimensional spaces, distances become less meaningful so reducing features, can help give more impact to the variance of features. Additionally, as LDA helps maximise the separation between classes and therefore should increase the accuracy of the distance based model. This is because increasing the distance between classes means that the nearest neighbours are more likely to be of the same class rather than another.

Additionally, using Mutual information classifier (MIC) on top of the LDA transformed data, can help identify features which have the highest dependency on the label's. Therefore, allowing us to further find features which provide higher information. As well as reducing irrelevant or noisy features. Enabling the training of the model to be more efficient and not overfit on irrelevant features. Fewer features also means fewer computations for the KNN model during training, reducing time and resources required.

Furthermore we can use GridSearchCV to find the most optimal hyperparameters for the model. In this case having $k=3$, $\text{weight}=\text{uniform}$, $\text{metric}=\text{manhattan}$ give the greatest accuracy.

3 Result Analysis and Visualisation

3.1 PERFORMANCE

Improved Implementation Model	
Performance Measure	Result
Accuracy	98%
F1-Score	98%
Recall	98%
Precision	98%

Figure 11: Improved KNN model results.

3.2 VISUALISATION

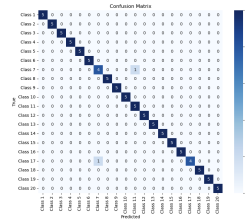


Figure 12: Improved KNN model Confusion Matrix with 50:50 training testing split.

3.3 ABLATION

The improvements made by the use the applying LDA as well as choosing the top 20 scored MIC categories does solve underlying issues with the data. Such as using LDA to increase separation between classes helped increase accuracy as well as enhance the discriminative variance. Additionally using MIC for feature selection helped reduce the grand dimensionality of the large data. As shown by figure 8, the number of features have reduced by 75%. However the accuracy suggests an overfitting nature resulting in a lack of generalisability.

4 Paradigm Design and Data Splitting

4.1 PARADIGM

To overcome the lack of generalisability, transitioning from KNN to an Ensemble model which utilises Support Vector Machines (SVM) as the base models can help solve this. While KNN is a simple and intuitive algorithm, it has shown to be sensitive and work less effective in very high dimensional data whilst being computationally inefficient. In contrast, SVMs are well adapted to handle high dimensional data, robust to overfitting, and capable of capturing complex, non-linear relationships through it's kernel. By combining SVMs with different kernels (RBF for brain data, Linear for text and image data), specialised models that excel in their respective domains can work together in an ensemble model.

Using the RBF kernel for non-linear data which have complex decision boundaries. The predictions of these models combine using majority voting, resulting in a balanced accurate and generalisable ensemble system.

4.2 ADJUSTMENT

We adjust the model by preprocessing the data via using LDA and MIC to choose the top 20 features to still retain a large amount of data. However we split the data into a 50:50 training to testing set split. This allows the model to train on more data making it so that the hyperplanes made in the SVM models are flexible. Therefore allowing the accuracy to be high. However this data split also enables the model to have enough data in test split to determine its generalisability. We can train each SVM model with their own chosen kernel on each different type of data separately. Resulting in the models to fit to each dataset separately, increasing the accuracy of using a specific data type to classify a label. Using these models in a voting system allows the high accuracy voting to happen between models to determine the labels.

4.3 REFLECTION

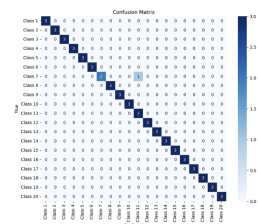


Figure 13: Improved KNN model Confusion Matrix with 70:30 train-testing split.

Implemented KNN Model	
Performance Measure	Result
Accuracy	98%
F1-Score	98%
Recall	98%
Precision	98%

Figure 14: Improved KNN model 70:30 split results.

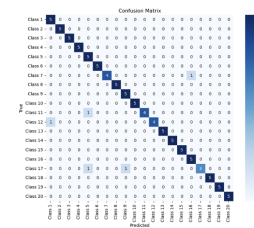


Figure 15: Ensemble model Confusion Matrix with 50:50 split.

Ensemble Model	
Performance Measure	Result
Accuracy	95%
F1-Score	95%
Recall	95%
Precision	96%

Figure 16: Ensemble model results.

Re-training and evaluating the model on the 50:50 split and ensemble paradigm reduces the accuracy by 3% but increases the generalisability. Transitioning from KNN to an ensemble of SVMs, combined with LDA and MIC preprocessing, reduced overfitting and enhanced performance. The 50:50 split provided balanced training and testing, ensuring powerful learning and evaluation whilst also preventing overfitting. The ensemble model, used SVMs with different kernels, outperformed KNN, displaying better classification across all classes. This approach highlights the importance of preprocessing, model selection, and data splitting in achieving optimal performance. Achieving a 96% precision score shows that our ensemble model is reliable in making fewer false positive error which is critical for reliable predictions. There is a decrease in the recall of the ensemble model compared to the improved KNN model using a 70:30 split, indicating that the ensemble model is less effective at capturing true positives and reducing the number of false negatives. Suggesting that the implemented KNN model using a 70:30 split is more capable to identify true positives than the ensemble model. On the other hand, the ensemble model is more generalisable in its predictions. Using the ensemble model is preferable as in practical applications, the ensemble model is more capable to determine an unseen object which it has a label for better than the implemented KNN model using a 70:30 and 50:50 split.

References

- [1] Changde Du, Kaicheng Fu, Jinpeng Li, and Huiguang He. Decoding visual neural representations by multimodal learning of brain-visual-linguistic features. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023. IEEE.
- [2] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, E. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [3] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, E. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.